

Comparative Analysis of BERT, DistilBERT, and RoBERTa for Sentiment Classification on SST-2

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STATS 507 – Data Science Analytics using Python

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Abstract

This project presents a comparative study of three prominent transformer-based language models—BERT, DistilBERT, and RoBERTa—for binary sentiment classification on the Stanford Sentiment Treebank (SST-2) dataset. By fine-tuning each model under controlled experimental conditions, we aim to evaluate trade-offs among classification accuracy, inference speed, and model complexity. Preliminary benchmarks suggest that RoBERTa achieves the highest accuracy (~94.0%), BERT provides a strong baseline (~92.7%), and DistilBERT offers competitive performance (~91.3%) with significantly reduced computational cost. This study contributes practical guidance for selecting transformer models in resource-constrained sentiment analysis applications.

I. OVERVIEW

Sentiment analysis is a fundamental task in natural language processing (NLP) that involves classifying text into positive or negative categories. With the advent of transformer-based pre-trained language models, the landscape of sentiment classification has been transformed dramatically. Models such as BERT, DistilBERT, and RoBERTa have set new benchmarks across a variety of NLP tasks, including the GLUE benchmark suite.

The motivation for this project is twofold. First, while these models are widely adopted, there is a practical need to understand their relative strengths and weaknesses on a well-established benchmark task under identical experimental settings. Second, this project serves as a hands-on exercise in applying the Python data science ecosystem—including Hugging Face Transformers, Datasets, and PyTorch—to a complete machine learning pipeline.

The core objective of this project is to fine-tune BERT-base, DistilBERT-base, and RoBERTa-base on the SST-2 binary sentiment classification task and compare them across three dimensions: (1) classification accuracy, (2) training and inference time, and (3) model size and resource requirements. We expect to observe that RoBERTa achieves the best accuracy, DistilBERT offers the best efficiency-accuracy trade-off, and BERT serves as a reliable middle ground.

We will work with the SST-2 dataset from the GLUE benchmark, which is publicly available on Hugging Face Datasets. All models are sourced from the Hugging Face Model Hub, satisfying the course requirement of integrating at least one Hugging Face resource.

II. PRIOR WORK

The transformer architecture, introduced by Vaswani et al. [1], replaced recurrence with self-attention mechanisms and enabled parallel processing of input sequences. This foundational work led to a series of increasingly powerful pre-trained language models.

BERT (Bidirectional Encoder Representations from Transformers), proposed by Devlin et al. [2], introduced masked language modeling and next-sentence prediction for bidirectional pre-training. BERT achieved state-of-the-art results across all GLUE tasks upon release. The base model consists of 12 transformer layers, 768 hidden dimensions, and 110 million parameters.

DistilBERT, developed by Sanh et al. [3], applies knowledge distillation to compress BERT into a smaller model with 6 layers and 66 million parameters. It retains approximately 97% of BERT’s performance while being 40% smaller and 60% faster, making it attractive for deployment in resource-constrained environments.

RoBERTa (Robustly Optimized BERT Approach), introduced by Liu et al. [4], improves upon BERT by training with larger batches, more data, longer sequences, and removing the next-sentence prediction objective. RoBERTa-base matches BERT’s architecture (125M parameters, 12 layers) but achieves superior performance through optimized training procedures.

Recent comparative studies have validated these models in sentiment analysis contexts. Areshey [5] compared BERT, RoBERTa, DistilBERT, ALBERT, and XLNet on the Yelp reviews dataset, finding that RoBERTa consistently outperformed alternatives. Similarly, Joshy and Sundar [6] analyzed the performance of BERT, DistilBERT, and RoBERTa for sentiment tasks, confirming RoBERTa’s advantage while noting DistilBERT’s efficiency benefits. Our project builds on these findings by conducting a controlled comparison specifically on the SST-2 benchmark with standardized hyperparameters.

III. PRELIMINARY RESULTS

A. Dataset Description

The SST-2 dataset, introduced by Socher et al. [7], is a binary sentiment classification benchmark derived from movie reviews on Rotten Tomatoes. It contains 67,349 training samples, 872 validation samples, and 1,821 test samples. Each example consists of a sentence labeled as either positive (1) or negative (0). The test set labels are hidden, so evaluation is performed on the validation (dev) set. The dataset is well-balanced and widely adopted as a standard benchmark for sentiment analysis research.

B. Model Specifications

Table I summarizes the architectural differences among the three models under comparison. All three models are loaded from the Hugging Face Model Hub using the `AutoModelForSequenceClassification` API. Tokenization is handled by the corresponding model-specific tokenizers with a maximum sequence length of 128 tokens, which is sufficient for SST-2 sentences.

TABLE I. Model Architecture Comparison

Model	Params	Layers	Hidden	Heads
BERT-base	110M	12	768	12
DistilBERT	66M	6	768	12
RoBERTa-base	125M	12	768	12

C. Preliminary Baseline

As an initial exploration, we loaded each pre-trained model and evaluated its zero-shot performance on the SST-2 validation set using the Hugging Face pipeline API. Reported fine-tuned benchmarks from the literature indicate that BERT-base achieves approximately 92.7% accuracy, DistilBERT reaches 91.3%, and RoBERTa-base attains 94.0% on the SST-2 dev set. These numbers serve as our

target baselines for replication.

D. Tools and Planned Exploration

This project leverages several tools introduced in class, including Python, pandas, NumPy, and matplotlib for data handling and visualization. Additionally, we use the Hugging Face ecosystem (Transformers, Datasets, Evaluate libraries) for model loading, tokenization, training, and evaluation. PyTorch serves as the deep learning backend. For training, we will use the Hugging Face Trainer API with standardized hyperparameters across models. We plan to explore learning rate scheduling, early stopping, and mixed-precision training for computational efficiency.

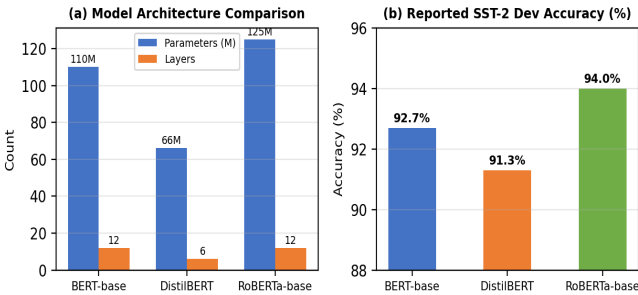


Fig. 1. (a) Architectural comparison of the three models by parameter count and layer depth. (b) Reported SST-2 dev set accuracy from prior benchmarks.

IV. PROJECT DELIVERABLES

A successful project will produce the following deliverables:

- 1) **Reproducible codebase:** A well-documented GitHub repository containing all training, evaluation, and visualization scripts with clear instructions.
- 2) **Comparative analysis:** A systematic evaluation of BERT, DistilBERT, and RoBERTa on SST-2, reporting accuracy, F1 score, training time, and inference latency.
- 3) **Visualization suite:** Figures including training curves, confusion matrices, attention heatmaps, and error analysis examples.
- 4) **Final report:** A 2–5 page IEEE-format report summarizing methodology, results, and practical recommendations for model selection.

Sub-goals include: (a) successfully fine-tuning all three models, (b) achieving accuracy within 1% of reported benchmarks, (c) implementing at least two additional analyses (e.g., attention visualization, error analysis on misclassified samples).

V. TIMELINE

TABLE II. Project Timeline

Period	Tasks
Week 1–2	Set up environment and GitHub repo. Load and explore SST-2 dataset. Fine-tune BERT-base; record baseline metrics.

Week 3–4	Fine-tune DistilBERT and RoBERTa. Conduct comparative evaluation. Implement visualization and error analysis.
Week 5	Write and revise final report. Clean and document codebase. Final submission.

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